Based on the decrease in count rate, 3 half-lives have passed. Hence half-life is 8 hours.

30 A For an alpha decay, nucleon number decreases by 4, proton number decreases by 2.
For a beta decay, nucleon number remains unchanged, proton number increases by 1.

Thus for 3 alpha decays and 1 beta decay, nucleon number will decrease by 12 and proton number will decrease by 5.